# CHAPTER 5 - TESTING, RESULTS & EVALUATION

## 5.1 Introduction

This chapter reports how TrackWIL was tested and evaluates the system against the requirements in Chapter 3. The purpose of testing was to show that every functional requirement works as specified, to measure the non-functional requirements wherever possible, and to find and fix defects before the final demonstration. The chapter describes the test strategy, summarises the test cases and their results, explains the defects that were found and corrected, presents the performance and usability evaluation, and evaluates each requirement. All test cases, defects and traceability links are recorded in the testing workbooks in the folder 06_Testing_Results_and_Evaluation.

## 5.2 Test Strategy

Testing followed the Test Plan and combined manual and automated testing at the system level.

- **Manual functional testing (TC-001 to TC-014).** Sihle Kayise tested the main user journeys in the browser on 22 October 2026: signing in and out as a student and as a coordinator, viewing and creating logbook entries, navigation, and invalid sign-in and empty-field validation.
- **Automated system testing (TC-110 to TC-144 and TC-150 to TC-175).** A test script (tests/system-tests.mjs, run with npm run test:system) sends the same requests as the browser to the running application and checks both the response and the database. It covers every functional requirement, the security rules, database constraints and response times, and can be repeated after every change. On 26 October 2026 it was extended with 26 tests for WIL placements, documents, the student profile and WIL completion, and for the retests of DEF-05 to DEF-07.
- **Browser and device checks (TC-145 to TC-146).** The main screens were checked in Google Chrome 155 and Microsoft Edge 155 at a phone width of 375 pixels and at desktop width.
- **Code and data reviews (TC-147 to TC-148).** Type checking, linting, the README set-up steps and the stored data were reviewed.
- **Usability testing with users (TC-149).** Planned, but blocked until written campus permission is received.

**Environment.** Windows 11 Pro; Node.js 24.21; TrackWIL production build (npm run build, npm start) with the SQLite database reset to the synthetic seed data before each run; Chrome 155.0 and Edge 155.0. No real personal data was used.

**Pass criteria.** A test passes when the actual result matches the expected result recorded in the workbook. Defects are logged, fixed, and retested with the same test case, and the full automated suite is rerun after every fix (regression testing).

## 5.3 Test Cases and Results

*Table 1: Summary of test results*

| Test group | Test cases | Requirements | Pass | Fail | Blocked |
|---|---|---|---|---|---|
| Manual functional tests (Sihle Kayise) | TC-001 to TC-014 | FR-01, FR-04, FR-09 to FR-11, FR-14 | 14 | 0 | 0 |
| Sign-in, sign-up and access control | TC-110 to TC-121 (incl. TC-120b) | FR-01 to FR-03, NFR-03 | 13 | 0 | 0 |
| Logbook entries and evidence | TC-122 to TC-126 | FR-04, FR-05, FR-13 | 5 | 0 | 0 |
| Verification links | TC-127 to TC-137 | FR-06 to FR-08 | 11 | 0 | 0 |
| Logbook, dashboard, timestamps and sign-out | TC-138 to TC-141 | FR-09 to FR-12, FR-14 | 4 | 0 | 0 |
| Security, integrity and performance | TC-142 to TC-144 | NFR-02 to NFR-04 | 3 | 0 | 0 |
| WIL placements | TC-150 to TC-156 | FR-03, FR-10, FR-15 | 7 | 0 | 0 |
| WIL documents | TC-157 to TC-165 | FR-03, FR-16, NFR-03 | 9 | 0 | 0 |
| Student profile and access control | TC-166 to TC-171 | FR-03, FR-11, FR-17, NFR-03 | 6 | 0 | 0 |
| WIL completion | TC-172 to TC-174 | FR-11, FR-18 | 3 | 0 | 0 |
| Sign-in page demo accounts | TC-175 | FR-01 | 1 | 0 | 0 |
| Browser, code and data checks | TC-145 to TC-148 | NFR-05 to NFR-07 | 4 | 0 | 0 |
| Usability with first-time students | TC-149 | NFR-01 | 0 | 0 | 1 |
| Total | 81 test cases | | 80 | 0 | 1 |

All 81 test cases are recorded in the Test Cases and Results workbook with their preconditions, steps, test data, expected and actual results, tester, date and evidence. The automated run logs (evidence/system_test_run_2026-10-23.txt and evidence/system_test_run_2026-10-26.txt) contain the actual output of every automated test. The final regression run on 26 October 2026, on Next.js 16.4.3, passed all 62 automated tests.

The results that matter most for the research aim are:

- **Account-free verification works end to end.** Submitting an entry produced a 48-character link that expires after exactly 7.00 days (TC-127). The mentor could open it without signing in (TC-130), approval set the entry to Verified with a timestamp (TC-131), and a rejected entry could be resubmitted with a new link (TC-132).
- **Links cannot be misused.** A used link (TC-133), an unknown link (TC-134) and an expired link (TC-135) were all refused without changing the entry, and two simultaneous approvals on the same link recorded only one decision (TC-136).
- **Access control holds.** Signed-out visitors, students and staff could reach only their own areas, and a blocked page sent none of its data (TC-117 to TC-120b, TC-171). Evidence files returned 200 to the owner and to a coordinator, 403 to another student, 401 to a signed-out visitor and 404 to a path-traversal attempt (TC-126).
- **Monitoring is accurate.** The dashboard showed Dylan as On track, Moe as Overdue and Sihle as Awaiting verification, matching the seed data (TC-139), and the logbook showed the correct verified hours (TC-138).
- **Placements, documents and completion work as specified.** Each placement counts its own verified hours (TC-155); documents are kept once per type and once per semester and are replaced on a new upload (TC-157 to TC-159); renamed, empty and oversized files are refused (TC-160); and a placement is marked WIL completed as soon as its required hours are verified, after which every change to it is refused (TC-172, TC-173).

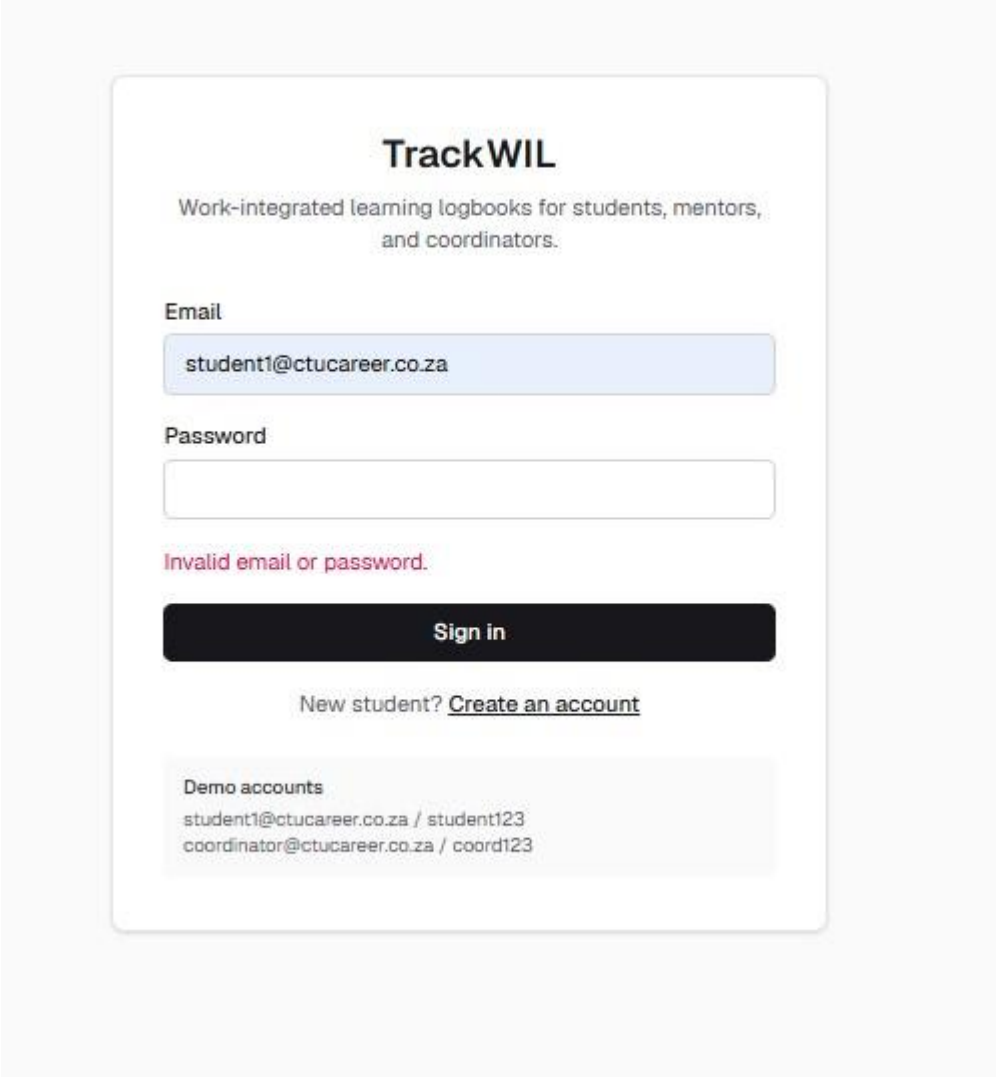



*Figure 1: Manual test TC-011: an invalid sign-in is refused with a clear message (Sihle Kayise, 22 October 2026)*

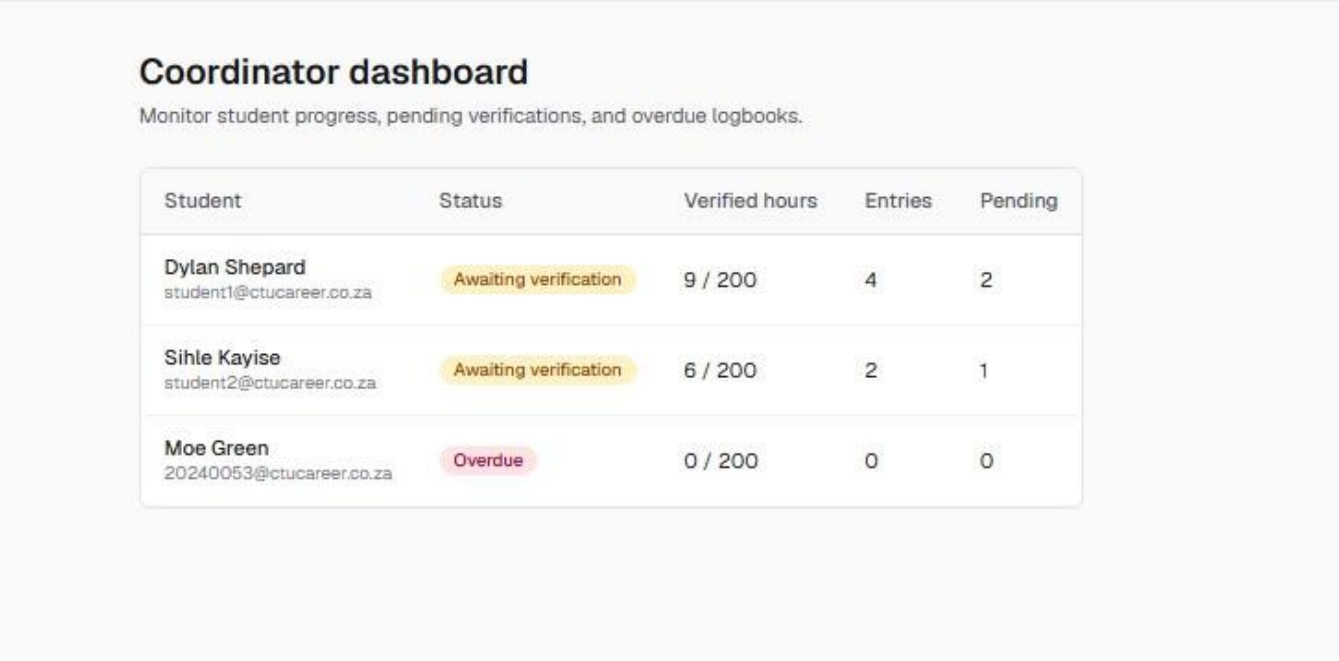



*Figure 2: Manual test TC-009: coordinator dashboard during manual testing (Sihle Kayise, 22 October 2026)*

## 5.4 Defects and Corrections

Seven defects were logged during testing. All seven were fixed, and each was retested with its own test case, followed by a full regression run on 26 October 2026.

*Table 2: Defects found, fixes and retest results*

| ID | Defect | Severity | Fix | Retest |
|---|---|---|---|---|
| DEF-01 | The session cookie held the user id without a signature. Setting the cookie to "coordinator-1" opened the coordinator dashboard without a password (reproduced on the earlier build: HTTP 200). | Critical | The cookie is now the user id plus an HMAC-SHA256 signature, checked in constant time; edited cookies are rejected. | Closed: TC-121 pass |
| DEF-02 | If a mentor link expired before the mentor responded, the entry stayed Pending verification and could never be resubmitted. | Medium | An entry with no active link can be resubmitted; the entry page shows an expiry notice. | Closed: TC-137 pass |
| DEF-03 | On a 375-pixel phone screen the logbook and dashboard tables widened the page to 580 pixels and cut off columns. | Medium | Full-width page container; tables scroll inside their box. | Closed: TC-145 pass (Chrome and Edge) |
| DEF-04 | The build reports a known security vulnerability in Next.js 16.4.0. | High | Upgraded Next.js to 16.4.3 (pull request #16) and reran the full test suite. | Closed: 62 automated tests pass; npm audit reports no vulnerabilities |
| DEF-05 | The sign-in page lists demo accounts but not the lecturer account. | Low | Every demo account, including the lecturer, is now listed. | Closed: TC-175 pass |
| DEF-06 | The new entry and edit placement pages showed an error screen, because database rows were passed to the browser in a form React does not accept. | Medium | The repository copies every row into a plain object; the timing test now also fails on any page error. | Closed: TC-156 pass |
| DEF-07 | A role check only in a layout redirected the browser, but Next.js still sent the page's data in the redirect response: a student, or anyone with a made-up session cookie, could read every student's name, email and hours. | Critical | Every protected page checks the role itself (requireRole) before loading any data. | Closed: TC-171 pass; TC-118, TC-119 and TC-121 now check the response body |

DEF-01 is the most important finding. It was found by reviewing the session code against the OWASP guidance used in Chapter 2, and it shows the value of testing security as well as functionality: every functional test passed before the fix, yet any user could have impersonated a coordinator. The defect was first reproduced on the previous build, then fixed and retested, and the automated suite now includes a test that forged and tampered cookies are rejected, so the problem cannot return unnoticed.

DEF-07 teaches a similar lesson. It was found while testing the student profile (FR-17), but it also affected the coordinator dashboard, which had passed every earlier test: those tests accepted any redirect as blocked without checking what the response contained. The access-control tests now check that the protected content is absent from the response, and they failed on the earlier build before the fix.

## 5.5 Performance / Usability Evaluation

**Performance (NFR-02).** Each main page was requested 10 times on the test computer with the demonstration data. Every response was far below the 2-second target.

*Table 3: Measured response times (TC-144, 26 October 2026)*

| Page or action | Average (ms) | Maximum (ms) | Target |
|---|---|---|---|
| Sign-in page (/) | 3 | 5 | under 2 000 ms |
| Student placements (/student) | 4 | 4 | under 2 000 ms |
| Placement logbook | 5 | 7 | under 2 000 ms |
| New entry form | 4 | 4 | under 2 000 ms |
| Coordinator dashboard | 5 | 7 | under 2 000 ms |
| Student profile | 5 | 5 | under 2 000 ms |
| Mentor verification page | 3 | 5 | under 2 000 ms |
| Create a logbook entry | 10 | 10 | under 2 000 ms |

These times were measured on the same computer as the server, so they show that the application and database are fast. They do not include internet delays, which would add time in a hosted deployment.

**Compatibility (NFR-06).** All main screens worked in Chrome 155 and Edge 155. At 375 pixels the first check found DEF-03; after the fix every screen fits the phone width, and the mentor verification page, which mentors are most likely to open on a phone, needs no sideways scrolling at all.

**Usability (NFR-01).** The interface was designed for quick use (Chapter 4, Section 4.5): an entry needs four fields, statuses use colored labels with text, and the mentor page has two buttons. However, the planned timed task with first-time students (TC-149) could not be run because written campus permission for testing with students has not yet been received. NFR-01 is therefore not yet proven by user data, and this is stated as a limitation.

## 5.6 Requirements Evaluation

*Table 4: Evaluation against the requirements*

| Requirement | Result | Evidence |
|---|---|---|
| FR-01 to FR-03: sign-in, sign-up, role-based access | Met | TC-001, TC-008, TC-011, TC-013, TC-110 to TC-121, TC-168, TC-171, TC-175 |
| FR-04 to FR-05: logbook entries and evidence | Met | TC-005, TC-006, TC-012, TC-014, TC-122 to TC-125 |
| FR-06 to FR-08: submission and one-time mentor links | Met | TC-127 to TC-137 |
| FR-09 to FR-11: statuses, logbook and monitoring dashboard | Met | TC-002 to TC-004, TC-009, TC-138, TC-139, TC-170, TC-174 |
| FR-12 to FR-14: timestamps, protected evidence, sign-out | Met | TC-007, TC-010, TC-126, TC-140, TC-141 |
| FR-15: WIL placements with their own required hours | Met | TC-150 to TC-156 |
| FR-16: WIL documents | Met | TC-157 to TC-165 |
| FR-17: student profile for coordinators and lecturers | Met | TC-166 to TC-170 |
| FR-18: WIL completion and locking | Met | TC-172 to TC-174 |
| NFR-01 Usability | Partially met: designed for, not yet measured with users | TC-149 blocked |
| NFR-02 Performance | Met (local measurement) | TC-144 |
| NFR-03 Security | Met after the DEF-01, DEF-04 and DEF-07 fixes | TC-121, TC-126, TC-127, TC-142, TC-160, TC-162, TC-171 |
| NFR-04 Reliability and integrity | Met | TC-136, TC-143 |
| NFR-05 Maintainability | Met | TC-147 |
| NFR-06 Compatibility | Met after DEF-03 fix | TC-145, TC-146 |
| NFR-07 Privacy | Met | TC-148 |

All eighteen functional requirements are met, and six of the seven non-functional requirements are met. The Requirements Traceability Matrix links every requirement to its design component, implementation files and test cases. The requirements deferred in Chapter 3 (mentor accounts, written mentor feedback, mentor access to evidence, a separate lecturer dashboard, coordinator review of documents, notifications and reports) were not part of version 1.0 and are carried into the recommendations in Chapter 6.

## 5.7 User / Stakeholder Feedback

No feedback has yet been collected from real students, mentors, coordinators or lecturers, because the Ethical Clearance Confirmation does not allow personal data to be collected before written campus permission is granted. The evaluation therefore relies on testing against the requirements and on the feedback that was available:

- **Supervisor feedback on the proposal (7 October 2026)** narrowed the scope to three core features, which allowed them to be completed and tested thoroughly.
- **Supervisor feedback on Week 2** asked the team to remove template formatting and fix scope inconsistencies; these changes were made in the revised chapters.
- **Internal review during testing led to improvements to the system: the signed session cookie (DEF-01), resubmission of expired links (DEF-02), the mobile layout (DEF-03) and, after the Week 4 submission, the role check on every page (DEF-07).**
- The team's review of the WIL process after the Week 4 submission added WIL placements, document uploads, a student profile for staff and WIL completion (FR-15 to FR-18).

If permission is granted before the final submission, a short task-based test and questionnaire will be run with volunteer students and staff, and the results will be added to this section.

## 5.8 Chapter Summary

TrackWIL was tested with 81 test cases: 14 manual tests, 62 automated system tests and 5 browser, code and data checks. Eighty passed and one, the usability test with students, is blocked until ethical permission is granted. Testing found seven defects, including two critical access-control vulnerabilities, and all seven were fixed and retested. All functional requirements and six of the seven non-functional requirements are met, and response times were far below the target. Within the limits of synthetic data and the absence of user feedback, the results show that TrackWIL supports the full WIL logbook workflow: recording daily work for each placement, keeping the WIL documents, account-free mentor verification and progress monitoring up to WIL completion. Chapter 6 draws conclusions from these results.